# Eigenphysics 2025-2026

## Example Sheet 1: Power Series and Polynomials

(1) **Checking Rodrigues' Formula for Hermite Polynomials:** Evaluate the first seven Hermite polynomials using the recursion relation derived in lectures,

$$H_n(x) = \sum_{k=0}^{n} a_k x^k \quad \text{where} \quad \frac{a_k}{a_{k+2}} = \frac{(k+1)(k+2)}{2(k-n)} \quad \text{and} \quad a_n = 2^n.$$

Check that the same results for the first seven Hermite polynomials are obtained from the Rodrigues' formula

$$H_n(x) = (-1)^n \exp(x^2) \frac{d^n}{dx^n} \exp(-x^2).$$

(2) **Legendre Polynomials:** Solve the Legendre equation

$$(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + n(n+1)y = 0$$

with a power series

$$y = \sum_{k=0}^{\infty} a_k x^k.$$

Show that the recursion relation is

$$\frac{a_{k+2}}{a_k} = \frac{(k-n)(k+n+1)}{(k+1)(k+2)},$$

and that the series will terminate at $k = n$ to leave a polynomial of order $n$. Given that the coefficient of $x^n$ is chosen by convention to be

$$a_n = \frac{1}{2^n}\binom{2n}{n},$$

write down the first five Legendre polynomials, $P_n(x)$. Note that these polynomials are important in the quantum theory of angular momentum.

(3) **Rodrigues' Formula for Legendre Polynomials:** The Rodrigues' formula for Legendre polynomials is

$$P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2-1)^n.$$

Use this formula to evaluate the first five Legendre polynomials, and show that the results are identical to those obtained in Q2.

(4) **Laguerre Polynomials:** Solve the Laguerre equation

$$x\frac{d^2y}{dx^2} + (1-x)\frac{dy}{dx} + ny = 0$$

with a power series

$$y = \sum_{k=0}^{\infty} a_k x^k.$$

Show that the recursion relation is

$$\frac{a_{k+1}}{a_k} = \frac{(k-n)}{(k+1)^2},$$

and that the series will terminate at $k = n$ to leave a polynomial of order $n$. Given that the coefficient of $x^n$ is chosen by convention to be $(-1)^n$, write down the first five Laguerre polynomials, $L_n(x)$. Note that these polynomials are important in the quantum theory of the hydrogen atom.

(5) **Rodrigues' Formula for Laguerre Polynomials:** The Rodrigues' formula for Laguerre polynomials is

$$L_n(x) = e^x \frac{d^n}{dx^n}(x^n e^{-x}).$$

Use this formula to evaluate the first five Laguerre polynomials, and show that the results are identical to those obtained in Q4.

(6) **Type I Chebyshev Polynomials:** If we use de Moivre's theorem to expand $\cos nx$, the result can always be expressed as a polynomial in $\cos x$ of degree $n$. It follows that we can define polynomials of degree $n$ in $x$,

$$T_n(x) = \cos\left[n \cos^{-1} x\right];$$

these are called Chebyshev polynomials of the first kind. Write down the first six of these using de Moivre's theorem. Show that $T_n(x)$ satisfies the differential equation

$$(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + n^2 y = 0.$$

Solve this equation with a power series

$$y = \sum_{k=0}^{\infty} a_k x^k.$$

and show that the recursion relation is

$$\frac{a_{k+2}}{a_k} = \frac{(k-n)(k+n)}{(k+1)(k+2)}.$$

(7) **Type II Chebyshev Polynomials:** If we use de Moivre's theorem to expand $\sin(n+1)x$, the result can always be written as $\sin x$ times by a polynomial in $\cos x$ of degree $n$. It follows that we can define polynomials of degree $n$ in $x$,

$$U_n(x) = \frac{\sin\left[(n+1)\cos^{-1}x\right]}{\sqrt{1-x^2}};$$

these are called Chebyshev polynomials of the second kind. Write down the first six of these using de Moivre's theorem. Show that $U_n(x)$ satisfies the differential equation

$$(1-x^2)\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + n(n+2)y = 0.$$

Solve this equation with a power series

$$y = \sum_{k=0}^{\infty} a_k x^k.$$

and show that the recursion relation is

$$\frac{a_{k+2}}{a_k} = \frac{(k-n)(k+n+2)}{(k+1)(k+2)}.$$

(8) **Proving Leibnitz' Theorem:** By using the product rule for differentiation repeatedly show that

$$\frac{d}{dx}[f(x)g(x)] = f'g + fg'$$
$$\frac{d^2}{dx^2}[f(x)g(x)] = f''g + 2f'g' + fg''$$
$$\frac{d^3}{dx^3}[f(x)g(x)] = f'''g + 3f''g' + 3f'g'' + fg'''$$

and work out the corresponding result for the 4th and 5th derivatives. We notice that the coefficients involved are just those from Pascal's triangle which leads to the formula

$$\frac{d^n}{dx^n}[f(x)g(x)] = \sum_{k=0}^{n} \binom{n}{k} f^{(n-k)}g^{(k)}.$$

Use induction to prove this theorem i.e. show that (i) it is true for $n = 1$; (ii) if we assume it is true for $n$, it is true for $n+1$. You will need the result that

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}.$$

(9) **An Explicit Formula for Laguerre Polynomials:** Use Leibnitz' theorem on the Rodrigues' formula for the Laguerre polynomial, $L_n(x)$, which is discussed in Q5, to show that

$$L_n(x) = \sum_{k=0}^{n} (-1)^k \frac{(n!)^2}{(n-k)!(k!)^2}\, x^k.$$